

# Configuring SFML

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*Computer Graphics I*  
*Spring 2013*

The following instructions can be used to setup the SFML libraries on your home computer. Instructions for Windows, Linux, and OSX are provided herein.

## Windows

### Environment:

- Windows 7 or XP
- Microsoft Visual Studio 2010 or Microsoft Visual Studio 2010 express C++

### Ingredients:

- Code skeleton for your language
  - cgl-line.zip is on the CG1 myCourses page
    - Rasterizer.cpp
    - Rasterizer.h
    - lineTest.cpp
    - simpleCanvas.cpp
    - simpleCanvas.h
- Appropriate libraries for your language, OS and assignment
  - SFML-1.6-2010 is available on the CG1 myCourses page
    - This is a precompiled binary of SFML for VS2010

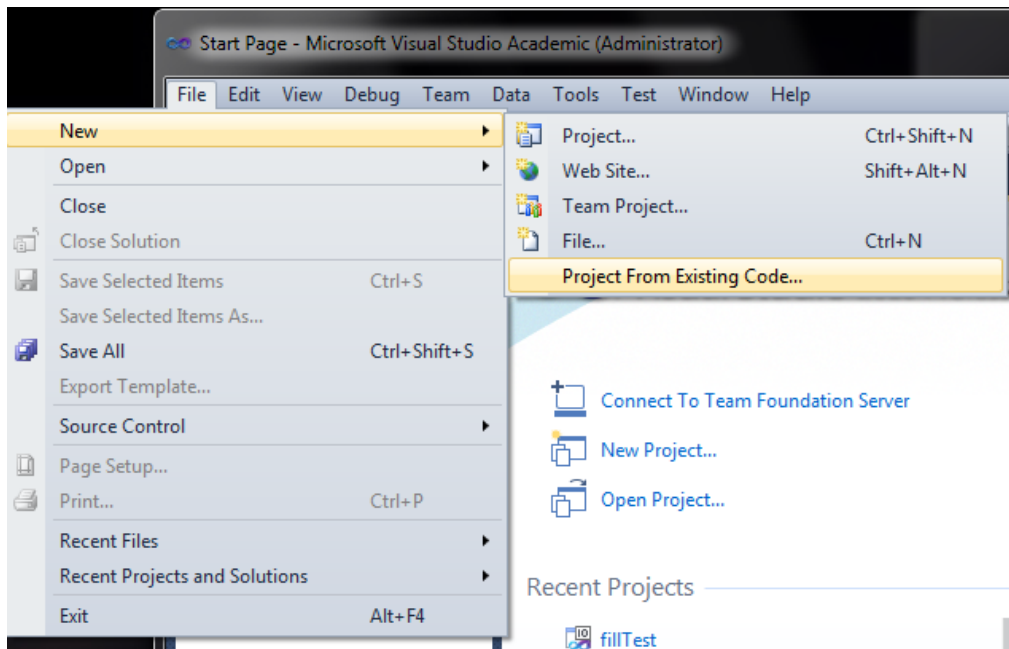
If you need a version other than Visual Studio 2010, downloads are provided on the SFML website, which is accessible from the assignment description.

### Steps:

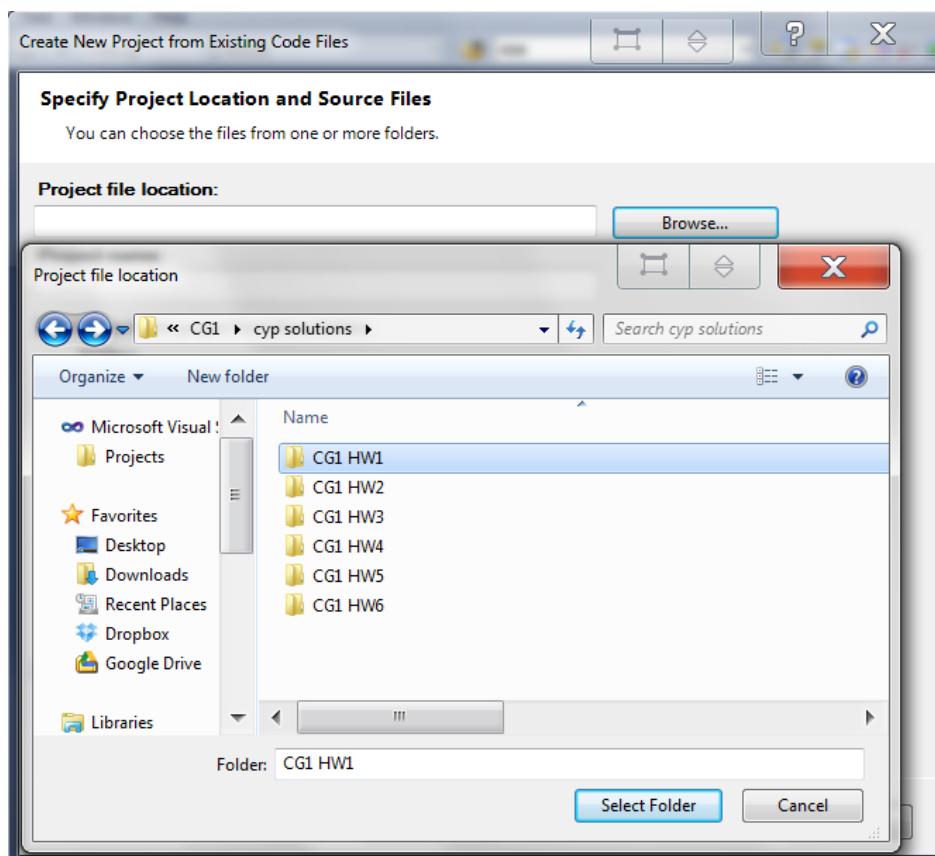
1. Copy SFML-1.6-2010\include\\* into C:\Program Files (x86)\Microsoft Visual Studio 10.0\VC\include\
2. Copy SFML-1.6-2010\lib\vc2010\\* into C:\Program Files (x86)\Microsoft Visual Studio 10.0\VC\lib\

\*\*\*NOTE\*\*\*: The path to the Visual Studio location may be different on your machines.

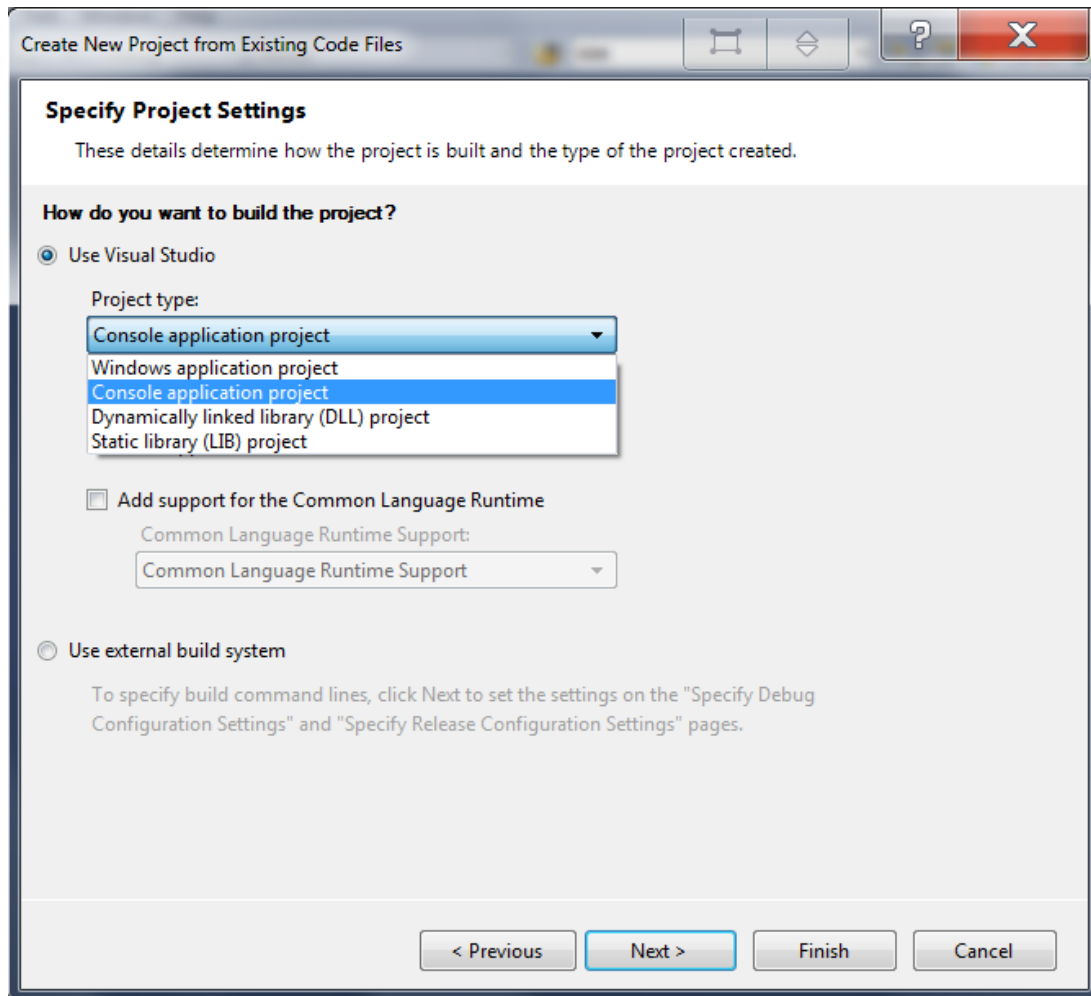
3. In Visual Studio 2010, make a new project from existing code.



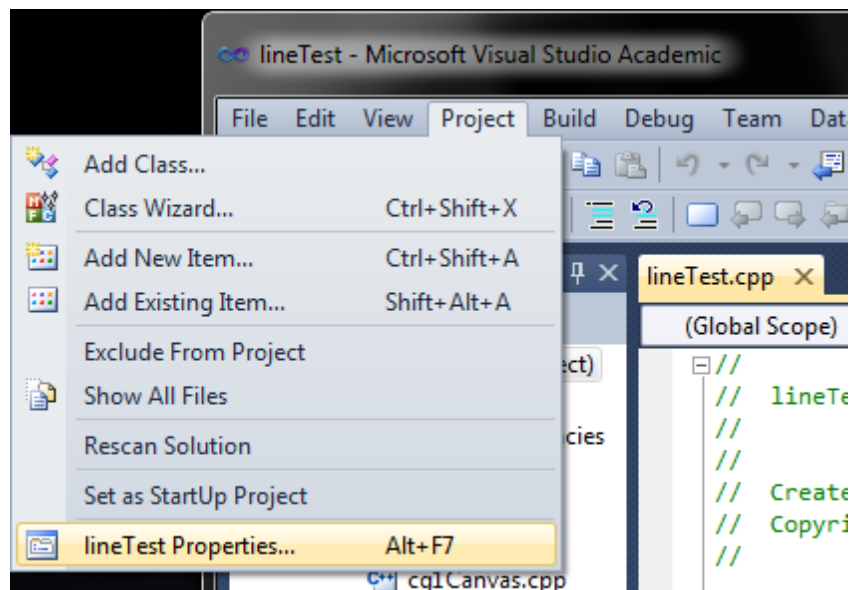
4. Make sure “Project file location” is the folder where you placed the assignment files.



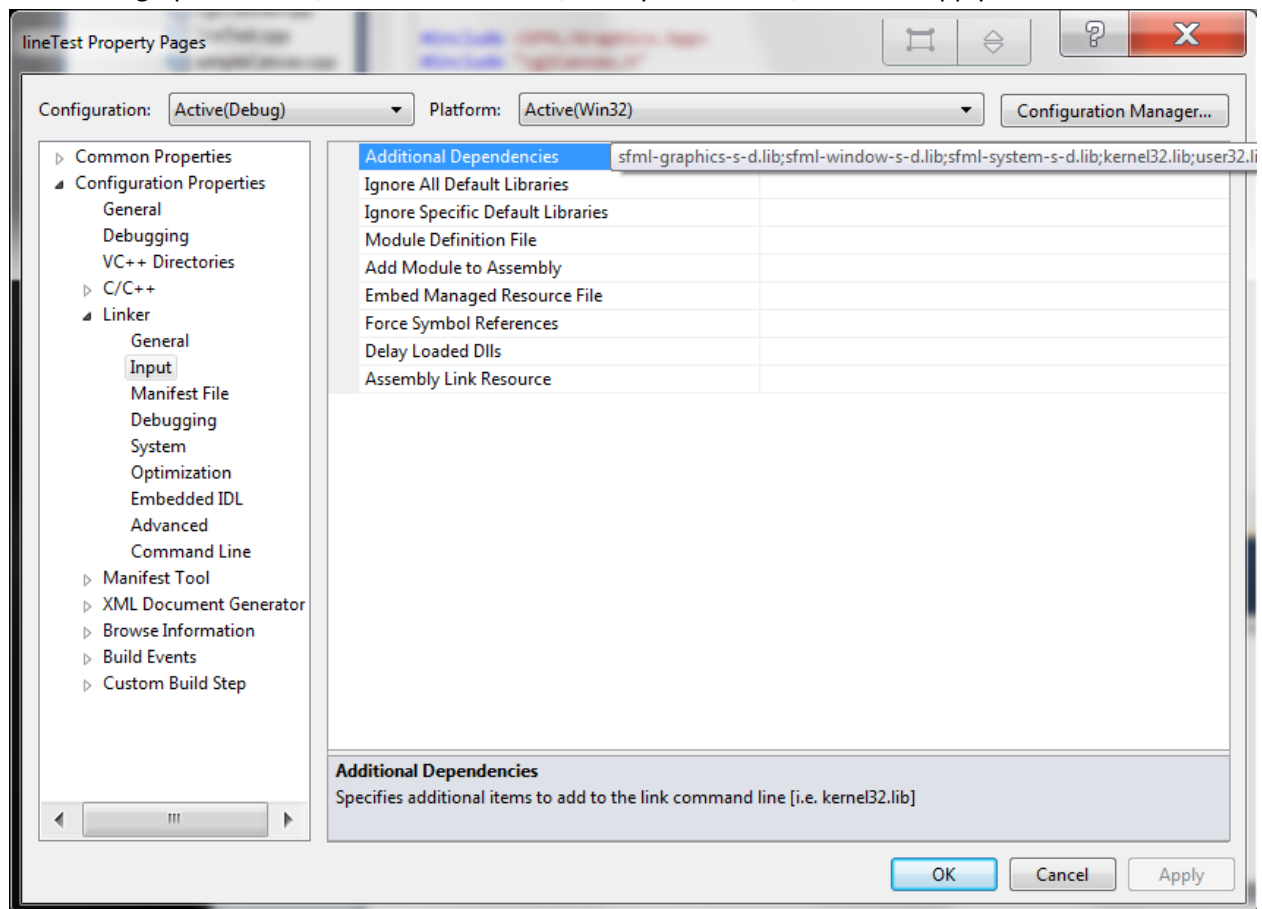
5. Make sure the “Project type” is “Console application project”.



6. With the project / solution open, from the "Project" menu, select "[project] Properties..."



7. To your “Configuration Properties” > “Linker” > “Input” > “Additional Dependencies”, Add “sfml-graphics-s-d.lib;sfml-window-s-d.lib;sfml-system-s-d.lib;” and hit “Apply”.



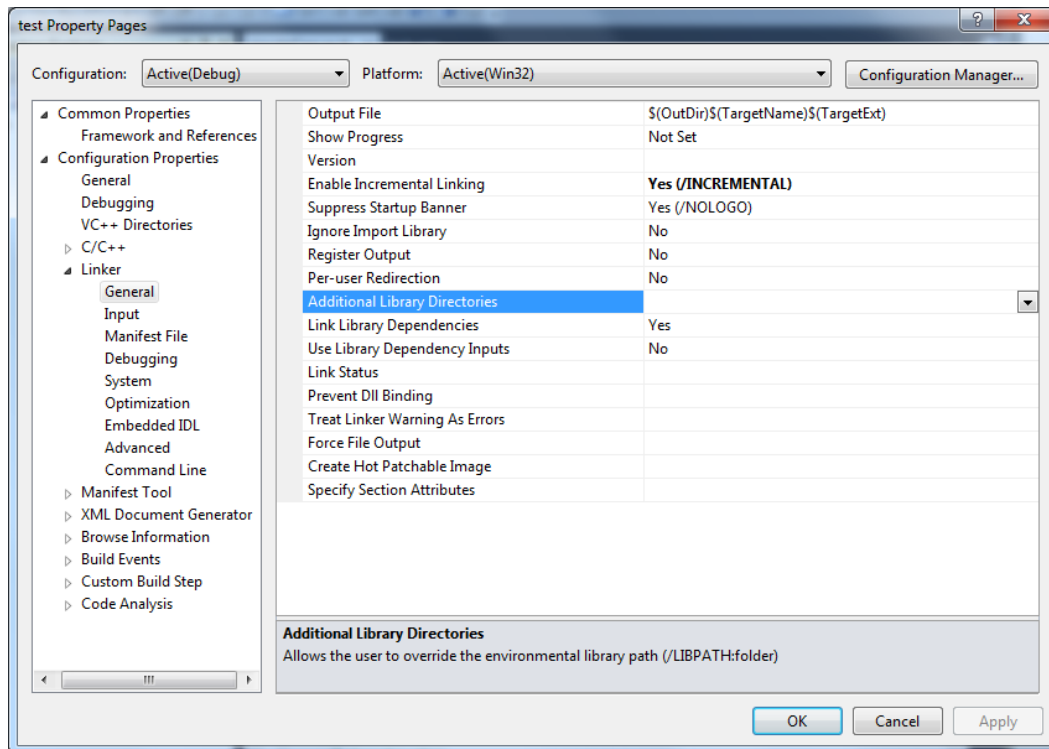
8. Now you should be able to run the skeleton project and get an empty window. When you build the project, you may see a number of warnings. Any warnings dealing with SFML libraries are OK. At the bottom of the output, it will indicate whether or not your code was successfully built.

## Important Windows Notes

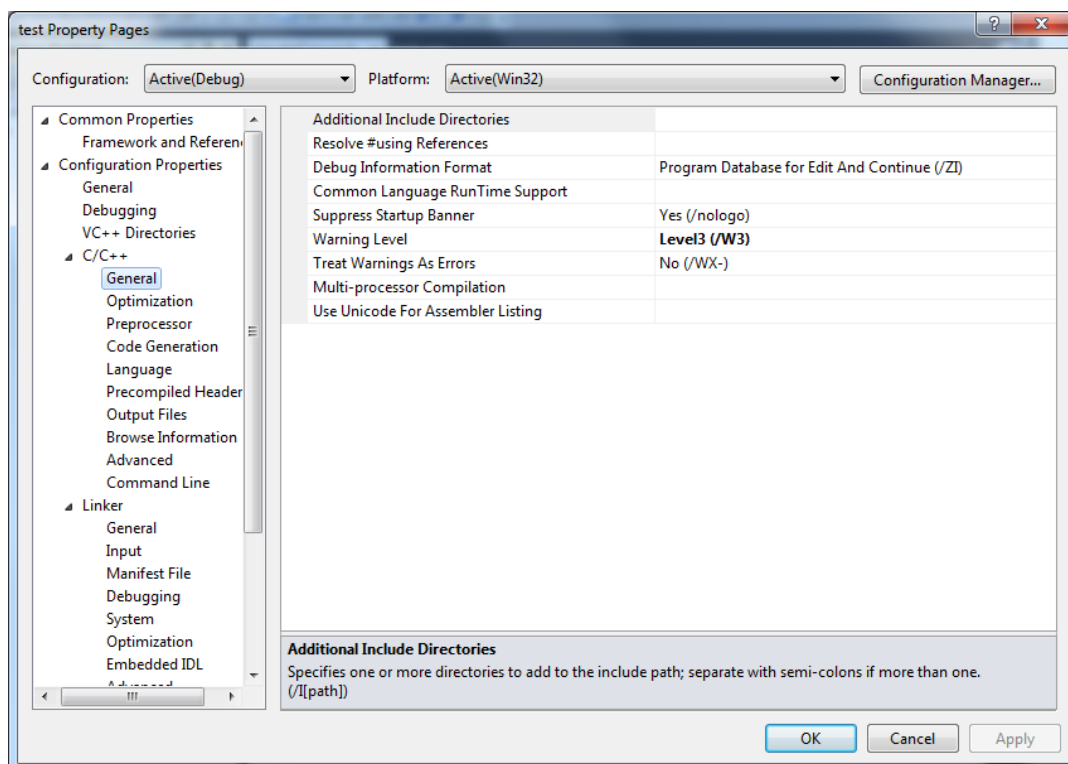
Steps 1 and 2 assume that you have administrator rights that allow you to copy/paste into the system directories. If you do not have these rights, there is another method that you can use.

After you add the additional libraries to the linker, you need to add the path to the lib and include directories of SFML. You may have the SFML download saved to any location...you will need to know that location in order to complete these steps.

Still under the Linker tab, select General. In the “Additional Library Directories”, you need to add the location to find the lib files. You may press the arrow and add a new folder in the dialog; this will allow you to use a file browser to select the location of the /lib directory. Otherwise, you may type the location in this box.



In addition to providing the library location, you need to provide the include path. This is done by selecting the C/C++ tab and then the General settings. You will need to add Additional Include Directories to point to the /include directory in your SFML download.



Once these steps are completed, your Visual Studio project is configured. This will have to be done to every project that gets created, as these settings are project specific.

## Linux

On the Linux machines in the CS labs the following commands should work (you should probably create a makefile for convenience):

```
g++ -c simpleCanvas.cpp
g++ -c Rasterizer.cpp
g++ -c lineTest.cpp
g++ -o test simpleCanvas.o Rasterizer.o lineTest.o -lsfml-system -lsfml-window -lsfml-graphics
./test
```

Assignment 2 and 3 are similar.

If you are doing Linux at home you will need to install SFML 1.6.

## OSX

(Instructions courtesy of Professor Strout)

It turns out the xcode project templates for SFML1.6 only work with the old xcode 3.x. So if you want to use xcode you should go the command line route and build a C++ application that uses the frameworks 'sfml-window' and 'sfml-graphics'. Here are the instructions:

To install SFML just follow the minimal install instructions from the website (under tutorials for 1.6). I list them out here for 64 bit mac's:

- Download and unpack SFML-1.6-dev-macosx-ub32+64.tar.bz2 to your download directory
- In a shell 'cd' to your download directory:

```
cd ~/Downloads
```

- Copy the files required for a minimal install into /Library/Frameworks

```
sudo cp -r SFML-1.6-dev-macosx-ub32+64/lib64/* /Library/Frameworks
sudo cp -r SFML-1.6-dev-macosx-ub32+64/extlibs/bin/x86_64/* /Library/Frameworks
```

From the command line you just need to change the link command:

```
g++ -c simpleCanvas.cpp
```

```
g++ -c Rasterizer.cpp
```

```
g++ -c lineTest.cpp
```

```
"g++ -o lineTest simpleCanvas.o Rasterizer.o lineTest.o -framework sfml-window -framework  
sfmlgraphics"
```

```
./lineTest
```